

Mark Scheme (Results)

Predicted Paper

Pearson Edexcel Level 3 Advanced GCE

in Economics B (9EB0)

Paper 02: Competing in the global economy

ADVANCED DIFFICULTY

General Marking Guidance

- All candidates must receive the same treatment. Examiners must mark the first candidate in exactly the same way as they mark the last.
- Mark schemes should be applied positively. Candidates must be rewarded for what they have shown they can do rather than penalised for omissions.
- Examiners should mark according to the mark scheme not according to their perception of where the grade boundaries may lie.
- There is no ceiling on achievement. All marks on the mark scheme should be used appropriately.
- All the marks on the mark scheme are designed to be awarded. Examiners should always award full marks if deserved, i.e. if the answer matches the mark scheme.
- Crossed out work should be marked UNLESS the candidate has replaced it with an alternative response.

Section A

Question 1(a)

Knowledge/understanding 1, Application 3

Knowledge/understanding: 1 mark for formula: Trade balance in goods and services = Exports – Imports (1).

Application: up to 3 marks: €1.93 trillion (1) – €1.71 trillion (1) = +€0.22 trillion / +€220 billion (1).

NB: If no working is shown: award 4 marks for +€0.22tn or +€220bn; 3 marks for 0.22 or 220 without units; zero if the sign is wrong.

(4 marks)

Question 1(b)

Knowledge/understanding 1, Application 2, Analysis 1

Knowledge/understanding: 1 mark for recognising that higher unit labour costs raise average costs of production, reducing cost competitiveness (1).

Application: up to 2 marks — e.g. German unit labour costs are ~40% above Spain and double those of Central European EU countries (1); high energy prices since 2022 add further cost pressure (1).

Analysis: 1 mark for developing the consequence — e.g. VW cannot undercut rivals such as BYD on price in China and is forced to cut margins or lose market share (1).

(4 marks)

Question 1(c)

Knowledge/understanding 1, Application 2, Analysis 1

Knowledge/understanding: 1 mark for identifying a feature of an export-led economy — e.g. large trade surplus or high exports-to-GDP ratio (1).

Application: up to 2 marks using Extract A — e.g. exports of €1.93tn against GDP of €4.12tn implies an export-to-GDP ratio of ~47% (1); current-account surplus of +5.9% of GDP indicates persistent external demand surplus (1).

Analysis: 1 mark — e.g. such a high exports-to-GDP ratio means growth and employment are closely tied to global trade conditions (1).

(4 marks)

Question 1(d)

Knowledge/understanding 2, Application 2, Analysis 2

Knowledge/understanding: up to 2 marks — e.g. the automotive sector is a significant share of German manufacturing and export earnings (1); loss of export revenues reduces aggregate demand via X falling (1).

Application: up to 2 marks — e.g. VW closing three German factories with up to 35,000 job losses (1); European brands' share of the Chinese EV market has fallen to 15% (1).

Analysis: up to 2 marks — e.g. negative multiplier effect through the supply chain: German component suppliers, engineers, and service firms lose demand (1); risk of hysteresis if high-skilled automotive workers cannot find comparable work (1).

(6 marks)

Question 1(e)

Knowledge/understanding 2, Application 2, Analysis 2, Evaluation 2

Candidates are expected to draw a long-run AS (LRAS) or PPF diagram, with a leftward shift/failure to shift right, compared with a counterfactual higher-investment path.

- Infrastructure (transport, energy, digital) is a key determinant of LRAS — raises productivity and reduces firms' costs.
- Persistent underinvestment leaves LRAS lower than it would otherwise be; potential output growth is slower.
- On a PPF, the frontier shifts outward more slowly relative to rival economies making stronger investment.
- Germany's gross fixed capital formation is 20.8% of GDP (Extract A); Extract C notes public investment has lagged peers for over a decade.
- Lower productivity raises unit labour costs relative to rivals, worsening competitiveness (consistent with VW's position in Extract B).
- Erosion of the export base over time shrinks the current-account surplus (currently +5.9% of GDP).
- Evaluation — private investment and FDI can partly offset weak public investment.
- Evaluation — modern economies can sustain output with existing infrastructure for a long time before effects become acute (lags matter).
- Evaluation — the effect depends on the initial stock of infrastructure; Germany's starting base is relatively strong.
- Evaluation — counterfactual matters: if rivals also underinvest, relative competitiveness may be less affected.

Level	Mark	Descriptor
0	0	A completely inaccurate response.
Level 1	1–2	Isolated elements of knowledge and understanding, using little or no relevant evidence. Arguments and chains of reasoning may be attempted. Limited attempt to address the question.
Level 2	3–5	Elements of knowledge and understanding, using limited relevant evidence. Arguments and chains of reasoning are developed. Judgements may be attempted.
Level 3	6–8	Accurate knowledge and understanding, supported throughout by use of relevant evidence which is well chosen. Arguments are developed, using logical, coherent chains of reasoning. A balanced awareness of competing arguments.

Demonstrating application (AO2): Where questions specifically stipulate the use of data or information provided in a stimulus, students must directly reference, interpret or analyse the information provided in the stimulus; in addition, they may select examples from their own knowledge but these must be relevant and directly connected to the context/issues set out in the stimulus. Where questions do not specifically stipulate the use of data or information provided in a stimulus, students must select relevant examples from their own knowledge, these must be directly connected to the context/issues set out in the question.

Question 1(f)

Knowledge/understanding 2, Application 2, Analysis 3, Evaluation 3

- Partnership gives VW faster access to Chinese EV technology (batteries, software, connected-car platforms).
- Cost-sharing reduces VW's R&D burden during an expensive industry transition.
- Local partners bring familiarity with Chinese consumer preferences and regulatory environment.
- Joint production may allow VW to benefit from Chinese economies of scale in battery production.
- Helps maintain access to the Chinese market at a moment when European brands hold only 15% of Chinese EV share.
- Evaluation — risks technology transfer: Chinese partners may become direct competitors in third markets.
- Evaluation — loss of strategic autonomy: VW becomes dependent on Chinese partners for critical components.
- Evaluation — geopolitical risk: future EU-China tensions could force VW to unwind partnerships at cost.
- Evaluation — profit-sharing dilutes VW's returns; the Chinese market has historically been VW's most profitable.
- Evaluation — reputational and workforce risks in Germany if partnerships are seen to hollow out domestic engineering capability.
- Evaluation — effectiveness depends on contract design, IP protections, and VW's ability to integrate Chinese technology into global product lines.
- Evaluation — overall: likely beneficial in the short run to protect Chinese revenues, but significant long-run strategic risks.

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Level 1	1–2	Isolated elements of knowledge and understanding, using little or no relevant evidence. Arguments and chains of reasoning may be attempted. Limited attempt to address the question.
Level 2	3–4	Elements of knowledge and understanding, using limited relevant evidence. Arguments and chains of reasoning are presented but with limited attempt to address the question. Comparisons and judgements may be attempted.
Level 3	5–7	Accurate knowledge and understanding, supported by use of relevant evidence to support the argument, clear chains of reasoning, well developed with arguments. An awareness of the significance of competing arguments is present although this may lack balance.
Level 4	8–10	Accurate knowledge and understanding, supported throughout by use of relevant evidence which is well chosen, logical, coherent chains of reasoning, showing full understanding of the question. Arguments are developed and evaluated. A full and balanced awareness of the validity and significance of competing arguments.

Demonstrating application (AO2): Where questions specifically stipulate the use of data or information provided in a stimulus, students must directly reference, interpret or analyse the information provided in the stimulus; in addition, they may select examples from their own knowledge but these must be relevant and directly connected to the context/issues set out in the stimulus. Where questions do not specifically stipulate the use of data or information provided in a stimulus, students must select relevant examples from their own knowledge, these must be directly connected to the context/issues set out in the question.

Question 1(g)

Knowledge/understanding 2, Application 2, Analysis 4, Evaluation 4

- €100bn over ten years = €10bn per year in additional public investment.
- Fiscal multiplier for infrastructure cited as 1.4–2.1 (Extract C) — each €1bn could add €1.4bn–€2.1bn to GDP in the medium term.

- Raises AD in the short run: G rises, creating jobs in construction and engineering (helpful given VW-related job losses).
- Shifts LRAS right in the long run through improved transport, digital and energy infrastructure.
- Higher productivity reduces unit labour costs, improving international competitiveness.
- Crowds in private investment: better transport and 5G networks raise the return on private capital.
- Green hydrogen and rail electrification support decarbonisation, aligning with EU climate obligations.
- Evaluation — financed by additional borrowing (requiring a change to the constitutional debt brake); raises the public debt trajectory.
- Evaluation — eurozone-wide interest-rate effects: higher German borrowing may raise Bund yields and transmit to other eurozone economies.
- Evaluation — crowding out: higher government demand for resources may bid up wages and materials prices.
- Evaluation — inflation risk given already-elevated HICP (5.9%); multipliers could be smaller if the economy is capacity-constrained.
- Evaluation — risk of poor project selection: multiplier depends critically on project quality (Extract C).
- Evaluation — political-economy risk: funds may be captured by politically connected regions or firms.
- Evaluation — ten-year horizon reduces short-run stimulus but improves sustainability; effectiveness depends on whether spending genuinely replaces private shortfalls or duplicates them.
- Evaluation — in the context of VW's restructuring and -0.3% growth in 2023, a countercyclical element is valuable.

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Level 2	3–5	Elements of knowledge and understanding, using limited relevant evidence. Arguments and chains of reasoning are presented but with limited attempt to address the question. Comparisons and judgements may be attempted.
Level 3	6–9	Accurate knowledge and understanding, supported by use of relevant evidence to support the argument, clear chains of reasoning, well developed with arguments. An awareness of the significance of competing arguments is present although this may lack balance.
Level 4	10–12	Accurate knowledge and understanding, supported throughout by use of relevant evidence which is well chosen, logical, coherent chains of reasoning, showing full understanding of the question. Arguments are developed and evaluated. A full and balanced awareness of the validity and significance of competing arguments.

Demonstrating application (AO2): Where questions specifically stipulate the use of data or information provided in a stimulus, students must directly reference, interpret or analyse the information provided in the stimulus; in addition, they may select examples from their own knowledge but these must be relevant and directly connected to the context/issues set out in the stimulus. Where questions do not specifically stipulate the use of data or information provided in a stimulus, students must select relevant examples from their own knowledge, these must be directly connected to the context/issues set out in the question.

Question 1(h)

Knowledge/understanding 2, Application 2, Analysis 4, Evaluation 4

- Structural supply-side reforms include labour-market flexibility, energy policy, training, deregulation, and innovation support.

- High unit labour costs (40% above Spain, per Extract B) reflect structural rigidities — reforms can improve competitiveness over time.
- Energy-price reductions (through supply diversification or renewables investment) directly address the 2022 energy-shock impact on VW.
- Skills training and R&D subsidies help close the EV/software capability gap identified in Extract B.
- Deregulation of product markets may speed adoption of new technology.
- Trade-policy reform (e.g. negotiating improved access to growing Asian markets) is partly a supply-side/structural issue.
- Evaluation — demand-side policy also plays a key role: infrastructure investment (see Extract C) is typically classified as fiscal/demand-side but has major supply-side effects.
- Evaluation — monetary policy can affect the euro's exchange rate, providing cyclical support to exporters (though the ECB targets eurozone inflation, not German competitiveness).
- Evaluation — supply-side reforms have long time lags; in the short run, demand-side measures provide essential cyclical support, especially given -0.3% growth.
- Evaluation — many of the symptoms of declining competitiveness (weak exports, job losses) have demand-side elements best addressed by countercyclical policy.
- Evaluation — political-economy constraints: structural reforms (job-protection rules, wage-bargaining) are highly contentious and slow to implement.
- Evaluation — supply-side and demand-side policies are complements rather than substitutes: the €100bn fund in Extract C operates on both dimensions.
- Evaluation — external factors (exchange rates, global demand, rivals' behaviour) may dominate policy choices.
- Evaluation — overall judgement: structural supply-side reform is necessary but not sufficient; a coherent policy mix is required.

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Level 3	6–9	Accurate knowledge and understanding, supported by use of relevant evidence to support the argument, clear chains of reasoning, well developed with arguments. An awareness of the significance of competing arguments is present although this may lack balance.
Level 4	10–12	Accurate knowledge and understanding, supported throughout by use of relevant evidence which is well chosen, logical, coherent chains of reasoning, showing full understanding of the question. Arguments are developed and evaluated. A full and balanced awareness of the validity and significance of competing arguments.

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Section B

Question 2

Knowledge/understanding 4, Application 4, Analysis 6, Evaluation 6

- Growth = increase in real GDP (or per capita); development = broader: health, education, income distribution, sustainability, freedoms.
- Rapid growth can fund development: higher tax revenues enable healthcare, schooling and infrastructure.
- India's GDP per capita of \$9,183 supports rising HDI (0.644), but inequality-adjusted HDI is much lower (0.475), suggesting gains are uneven.
- Indonesia shows similar pattern: GDP per capita \$14,687; HDI 0.713; inequality-adjusted HDI 0.584.
- Ecological footprint matters: India's per capita footprint (1.2 gha) is still below one-Earth sustainability (~1.6 gha), but rapid industrialisation is pushing it upward.
- Ethiopia's gross national savings at 33.1% of GDP provides potential for future investment-led growth.
- Brazil (GDP per capita \$17,827) has relatively high HDI but low savings rate (17.4%), limiting future investment.
- Growth can be exclusionary: capital-intensive export industries may enrich elites while leaving rural or informal-sector workers behind.
- Environmental costs: pollution, deforestation, biodiversity loss harm long-run welfare and resource sustainability.
- Evaluation — incompatibility claim is too strong: Germany's trajectory demonstrates growth can coexist with high HDI (0.950) and moderate inequality-adjusted HDI (0.891).
- Evaluation — Germany's ecological footprint (4.7 gha) shows the environmental cost of development: replicating this path globally is not sustainable.
- Evaluation — the issue is less whether growth and development are compatible and more the type of growth: inclusive, green, institution-building growth supports development; extractive, unequal, pollutive growth undermines it.
- Evaluation — Kuznets curve: inequality may rise then fall with development; China/South Korea examples show rapid catch-up without permanent incompatibility.
- Evaluation — environmental Kuznets curve: emissions per capita may rise then fall, but climate scientists warn developing economies cannot afford the same trajectory.
- Evaluation — data quality in developing economies is often weaker; cross-country comparisons require caution.
- Evaluation — balanced conclusion: rapid growth is not inherently incompatible with development, but without deliberate policy on distribution, environment and institutions, growth can undermine broader welfare improvements.

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Level 2	5–9	Elements of knowledge and understanding, using limited relevant evidence. Arguments and chains of reasoning are presented, but connections between causes and consequences are incomplete. Limited attempt to address the question. Comparisons, judgements or conclusions may be attempted, but are unsupported or generic.
Level 3	10–15	Accurate knowledge and understanding, supported by use of relevant evidence to support the argument, developed chains of reasoning, showing understanding of connections between causes and consequences. Arguments are well developed and competing arguments are present although this may lack balance. A conclusion may be attempted but may not show awareness of the significance of competing arguments.
Level 4	16–20	Accurate knowledge and understanding, supported throughout by use of relevant evidence which is well chosen and fully integrated to support the argument, well developed and logical, coherent chains of reasoning, showing full understanding of the question. Arguments are fully developed and evaluated. A full awareness of the validity and significance of competing arguments, leading to nuanced and balanced comparisons, judgements or conclusions.

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Section C

Question 3

Knowledge/understanding 4, Application 4, Analysis 6, Evaluation 6

- Industrial policy typically involves subsidies, tax credits, guaranteed procurement and other sector-specific support.
- TSMC receives substantial direct transfers: \$6.6bn in Arizona; €5bn in Dresden; ¥900bn in Kumamoto — reducing its effective cost of capital.
- Subsidies offset TSMC's admission that overseas fabs cost 50% more to operate than its Taiwanese facilities.
- TSMC's leading-edge process development remains in Taiwan — the firm retains strategic know-how and captures future upside.
- Long-term contracts with customers (Apple, NVIDIA, AMD) mean TSMC's revenue stream is largely guaranteed; subsidies effectively de-risk expansion.
- Financial accounting: subsidies flow directly through to retained earnings / investment capacity, strengthening TSMC's balance sheet.
- Economies of scale: TSMC's expanded capacity reduces average costs globally, benefiting the firm even before subsidies.
- Evaluation — host-economy benefits include jobs (Arizona, Dresden, Kumamoto create thousands of high-skilled jobs directly).
- Evaluation — supply-chain security: reliable domestic chip supply reduces the economic cost of any future Taiwan-related disruption.
- Evaluation — knowledge spillovers: local R&D capacity and supplier ecosystems can develop around TSMC fabs.
- Evaluation — TSMC states subsidies are 'necessary, not sufficient'; implies host governments are capturing genuine externalities that justify support.
- Evaluation — however, multiple governments subsidising overlapping capacity risks over-investment and a future glut.
- Evaluation — WTO records of >3,000 trade-distorting measures per year raise systemic risk of retaliation and fragmentation.
- Evaluation — opportunity cost: \$6.6bn in US CHIPS funding could have been used for education, healthcare, or basic research.
- Evaluation — 'picking winners' historically has mixed results; governments may back incumbents rather than innovators.
- Evaluation — who benefits depends on counterfactual: if TSMC would not have expanded without subsidies, governments may still have net benefit; if it would have invested anyway, subsidies are pure transfer.
- Evaluation — political economy: TSMC's bargaining power (supplying 90% of advanced chips) is so strong that subsidies are unlikely to fully capture the firm's willingness to invest.
- Evaluation — balanced conclusion: in the short run, TSMC is likely the clearer beneficiary; host economies gain strategic resilience and jobs but also bear significant fiscal costs and risk over-investment. The long-run verdict depends on whether chip demand grows fast enough to justify duplicated capacity and whether knowledge spillovers materialise.

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